

## Practical 11

Write Shell Script for followings :

A. To find the global complete path for any file.

```
student@student-virtual-machine:~/Downloads$ nano find_path.sh
student@student-virtual-machine:~/Downloads$ touch key_document.txt
student@student-virtual-machine:~/Downloads$ bash find_path.sh
Enter the file name: key_document.txt
Global complete path: /home/student/Downloads/key_document.txt
student@student-virtual-machine:~/Downloads$
```

```
GNU nano 6.2 find_path.sh
#!/bin/bash

read -p "Enter the file name: " filename

# Check if file exists and find its path
if [ -f "$filename" ]; then
    global_path=$(realpath "$filename")
    echo "Global complete path: $global_path"
else
    echo "File '$filename' not found in the current directory."
fi
```

B. To broadcast a message to a specified user or a group of users logged on any terminal.

```
student@student-virtual-machine:~/Downloads$ nano broadcast_msg.sh
student@student-virtual-machine:~/Downloads$ bash broadcast_msg.sh
Target user or 'all': all
Enter your message: The server will be down for maintenance at 3 PM for 30 minutes.
Please save your work.
Confirmation (yes/no): yes
Broadcasting message to all logged-on users...
Done.
```

```
GNU nano 6.2 broadcast_msg.sh
#!/bin/bash
# Broadcast Message Script
echo -n "Target user or 'all': "
read TARGET
echo -n "Enter your message: "
read MESSAGE
echo -n "Confirmation (yes/no): "
read CONFIRM
if [[ "$CONFIRM" == "yes" ]]; then
    if [[ "$TARGET" == "all" ]]; then
        wall "$MESSAGE"
        echo "Broadcasting message to all logged-on users... Done."
    else
        # Check if user is logged on
        if who | grep -q "^$TARGET"; then
            echo "$MESSAGE" | write "$TARGET"
            echo "Broadcasting message to user '$TARGET'... Done."
        else
            echo "User '$TARGET' is not currently logged on."
        fi
    fi
else
    echo "Broadcast cancelled."
fi
```

C. To copy the file system from two directories to a new directory in such a way that only the latest file is copied in case there are common files in both the directories.

```
student@student-virtual-machine:~/Downloads$ nano copy_latest.sh
student@student-virtual-machine:~/Downloads$ bash copy_latest.sh dir1 dir2 dest_dir
Setting up environment...
Comparing files and copying the latest versions...
Processing common file: file1.txt
    Newer version in dir2, copying.
Unique file in dir1: file2.txt, copying.
Unique file in dir2: file3.txt, copying.
File copy operation complete.
student@student-virtual-machine:~/Downloads$ ls -l dest_dir
total 12
-rw-r--r-- 1 student student 18 Jan  2 00:00 file1.txt
-rw-r--r-- 1 student student 19 Jan  3 00:00 file2.txt
-rw-r--r-- 1 student student 18 Jan  4 00:00 file3.txt
```

```
GNU nano 6.2 copy_latest.sh
#!/bin/bash
# Shebang

# Set directories from arguments
dir1=$1; dir2=$2; dest=$3

# Error check for arguments
if [ "$#" -ne 3 ]; then
    echo "Usage: $0 dir1 dir2 dest_dir"
    exit 1
fi

# Create dest directory if it doesn't exist
mkdir -p "$dest"

echo "Setting up environment..."
echo "Comparing files and copying the latest versions..."

# Loop through files in dir1
for file1 in "$dir1"/*; do
    fname=$(basename "$file1")
    if [ -e "$dir2/$fname" ]; then
        # File exists in both directories, compare modification time
        if [ "$dir2/$fname" -nt "$file1" ]; then
            echo "Processing common file: $fname"
            echo "    Newer version in dir2, copying."
            cp "$dir2/$fname" "$dest/"
        else
            echo "Processing common file: $fname"
            echo "    Newer version in dir1, keeping existing copy."
        fi
    else
        # Unique file in dir1
        echo "Unique file in dir1: $fname, copying."
        cp "$file1" "$dest/"
    fi
done

# Now process files unique to dir2
for file2 in "$dir2"/*; do
    fname=$(basename "$file2")
    if [ ! -e "$dest/$fname" ]; then
        # File not already in dest, so it must be unique to dir2
        echo "Unique file in dir2: $fname, copying."
        cp "$file2" "$dest/"
    fi
done
echo "File copy operation complete." ■
```